MC 451 Describing Data in R rubric

75 points. Due Friday, November 20, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 75. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Submissions are unlimited before the deadline, so you can revise and submit again.

- The calls these tables and figures lean on, all of which run the same in VS Code: `count()` for a frequency table, `pivot_wider()` to lay a cross-tabulation out, `fct_reorder()` to order the bars, `labs()` for the axis labels, title, and caption, `scale_fill_brewer()` or the v2v palette for the fill, and position = "fill" with `scales::percent_format()` for a proportion chart.
- Recommended, not graded here: open the script with the settings block Inferencing Data requires (load the data, rename the variables you use to a fixed set, and make the categorical ones factors), so that next week's script is this one with four tests added.
- What it feeds: your White Paper, which runs these same table and figure calls on your own variables. The Twitch figures stay here; the paper shows only what your own data shows.

Describing Data in R, 75 points

Criterion	Exemplary	Proficient	Developing	Beginning
Frequency tables 15 points	**15 points.** A frequency table for each key categorical variable, built with `count()` and labeled so a reader can tell which variable each one shows.	**11 to 14 points.** Tables for most key variables, or a table without labels.	**8 to 10 points.** One table, or tables printed as raw R output with no labels.	**0 to 7 points.** None.
The three numbers 15 points	**15 points.** Mean, median, and standard deviation for every continuous variable you report, and a sentence whenever the mean and median disagree.	**11 to 14 points.** All three numbers reported, but a large gap between the mean and the median goes unremarked. When they part company, the distribution is telling you something.	**8 to 10 points.** The mean alone, or the three numbers for only some variables.	**0 to 7 points.** No summary numbers.
Figures 20 points	**20 points.** At least two ggplot2 figures plus one figure of your own choice, each with axis labels, a title, and a caption, saved to a file by the code that makes them, with one sentence per figure disclosing a choice you made (a bin width, a cap on an axis, a category you merged), and one sentence on why you chose the third figure for that variable.	**15 to 19 points.** The figures are there, but one is missing a label, the title, the caption, or its disclosure sentence, or the figure you chose has no reason given.	**10 to 14 points.** One figure, or figures saved by hand instead of by the code.	**0 to 9 points.** No figures.

Criterion	Exemplary	Proficient	Developing	Beginning
Cross-tabulation 10 points	**10 points.** A cross-tabulation of two categorical variables, laid out with `count()` and `pivot_wider()`, labeled so a reader can tell the rows from the columns. This is the table next week's chi-square tests.	**8 to 9 points.** Present, but the labels are missing.	**5 to 7 points.** Built on the wrong columns (a variable crossed with itself, or an ID column).	**0 to 4 points.** Missing.
Alt text 5 points	**5 points.** Every figure has alt text that says what the figure shows, not a repeat of its caption.	**4 points.** Alt text on every figure, but one repeats the caption.	**3 points.** Alt text on some figures only.	**0 to 2 points.** None.
Interpretation 10 points	**10 points.** Two to three sentences after each output, saying what it shows and what it does not.	**8 to 9 points.** Sentences after each output, but they describe the output without saying what it cannot show.	**5 to 7 points.** Interpretation for some outputs only, or a single line each, which usually means the "what it does not show" half is missing.	**0 to 4 points.** None.